Intelligent Forest Fire Risk Assessment and UAV Dispatch Using Mamdani Fuzzy Logic in NVIDIA Isaac Sim

CSC3034 Computational Intelligence — Final Technical Project Report

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Abstract

Wildfires pose catastrophic ecological and atmospheric hazards, destroying terrestrial ecosystems and releasing stored carbon back into the atmosphere. Real-time fire risk assessment is difficult because the four variables that govern ignition — ambient temperature, relative humidity, localized wind velocity and fuel moisture — interact non-linearly, and no single one of them ranks two conditions correctly. Traditional binary alarm systems therefore fire on benign summer heat and stay silent when moderate conditions are compounded by high winds and drought-stressed undergrowth.

This paper proposes an explainable **Mamdani Fuzzy Logic Inference System** coupled with **NVIDIA Isaac Sim (OpenUSD)** for dynamic forest fire risk estimation and autonomous unmanned aerial vehicle (UAV) surveillance response. The fuzzy controller maps four continuous environmental inputs into three defuzzified consequents: a normalized **Fire Risk Index (FRI)**, an autonomous **UAV Patrol Urgency Throttle**, and a continuous **Suppression Alert Level** on [0, 3] whose tiers (Advisory, Watch, Warning, Evacuate/Suppress) are inferred by the rule base rather than thresholded from the risk index. A procedural 3D forest simulation stage was created in Isaac Sim containing forest terrain, varied vegetation moisture distributions, thermal wildfire hotspots, and an animated surveillance drone. Through a 12-waypoint simulated flight mission (seed 0, so the log is reproducible), the controller moved from benign patrol (20.1% urgency, FRI=45.0, alert 1.000) to emergency dispatch (89.5% urgency, FRI=91.4, alert 2.826) within two waypoints of entering a hotspot radius; the largest throttle change between consecutive waypoints, 69.8 points, occurs only on the step that leaves the second hazard radius. The inference engine was validated in two ways: 25 automated unit tests covering the scenario envelope, input boundaries and all three consequents, plus a per-scenario cross-check against the scikit-fuzzy reference implementation (worst deviation 0.069 across 9 scenarios). The three defuzzification algorithms were also benchmarked, confirming **Centroid of Area** as the choice where output continuity matters. The system is presented in alignment with **United Nations Sustainable Development Goals 13 (Climate Action)** and **15 (Life on Land)**.

Keywords: Computational Intelligence, Mamdani Fuzzy Inference, Wildfire Risk Modeling, Autonomous UAV Surveillance, NVIDIA Isaac Sim, OpenUSD, Sustainable Development Goals. ---

1. Introduction

1.1 Background & Motivation

Wildfires represent one of the most volatile and destructive natural hazards globally, and longer fire seasons have raised ignition frequency. Detection latency dominates containment cost: the earlier a hotspot is reported, the smaller — and cheaper — the response it needs. The design target adopted here is therefore a continuous risk estimate that updates at patrol cadence, rather than a single threshold that fires once.

However, remote forest telemetry arrives as four scalars whose effect on ignition is not additive, and no single threshold ranks them correctly:

- A high ambient temperature (38°C) in a tropical rain forest with 85% relative humidity poses low immediate combustion hazard.
- Conversely, a moderate temperature (26°C) accompanied by 15% relative humidity, severe wind gusts (65 km / h), and desiccated brush represents a critical firestorm risk.

The two readings differ by 12°C, yet the cooler one is the dangerous one: run through the controller below, the 38°C / 85% case scores FRI=14.0 (tier `NORMAL`) and the 26°C / 15% / 65 km / h case scores FRI=71.4 (tier `WARNING`), so any rule keyed on a single temperature threshold inverts the pair. The simulator reproduces this by construction: baseline conditions of 28°C, 48% RH, 25 km / h and 40% fuel dryness are displaced by hotspot proximity, adding up to 16°C, removing up to 32 percentage points of humidity, adding 20 km / h of wind and 50 points of fuel dryness inside a 12 m radius, with per-waypoint sensor jitter of ±1.0 to 1.5°C, ±2% RH, ±3 to 4 km / h and ±1 to 2 points.

Binary threshold triggers (e.g., if Temp > 40°C then sound alarm) fail to capture these interdependent gradients, causing excessive false alarms or missed early-stage hazards. Fuzzy set theory provides the paradigm for this problem: it replaces the comparison against a constant with a membership function, so continuous linguistic uncertainty can be carried through the inference instead of being collapsed at the input [2].

1.2 Alignment with UN Sustainable Development Goals (SDGs)

This research is directly aligned with two United Nations Sustainable Development Goals:

1. **SDG 13: Climate Action (Target 13.1)**: Minimizes massive greenhouse gas emissions produced by uncontrolled wildfires through proactive early aerial detection [6].
2. **SDG 15: Life on Land (Target 15.2)**: Preserves forest ecosystems, protects endangered wildlife habitats, and combats soil degradation.

1.3 Contributions & Objectives

The key objectives of this project are:

1. To formulate a multi-variable Mamdani fuzzy inference system with four environmental inputs and three continuous control outputs.
2. To construct an OpenUSD procedural forest simulation in NVIDIA Isaac Sim equipped with realistic terrain, varied fuel vegetation, and dynamic wildfire hotspots.
3. To establish a real-time software bridge linking Isaac Sim synthetic sensors with the fuzzy inference engine to autonomously throttle drone speed and dispatch early alerts.

4. To evaluate the mathematical stability and defuzzification characteristics of the controller across diverse environmental trajectories.

2. Literature Review

2.1 Fire Danger Rating: Indices Versus Linguistic Variables

Deterministic fire danger ratings dominate operational practice. The McArthur Forest Fire Danger Index fuses temperature, humidity, wind speed and a drought factor into one continuous score, and its category boundaries still anchor Australian fire-weather warnings [3]. The index is transparent to a duty officer, yet it is calibrated to eucalypt fuel and its warning tiers are hard cuts: a score of 49.9 and a score of 50.1 fall into different categories although the underlying hazard is indistinguishable. Exactly that discontinuity motivates the fuzzy treatment adopted here, in which a reading belongs to several linguistic classes at once with graded membership rather than to one class by comparison [2].

2.2 Mamdani Inference and the Defuzzification Choice

The linguistic controller of Mamdani and Assilian established the architecture reused in this project: fuzzify each antecedent, evaluate a conjunctive rule base with min, aggregate the clipped consequents with max, and defuzzify the union into one crisp actuation value [1]. Their formulation deliberately keeps the rule base readable in domain language, which is the property this work relies on for the explanation in Section 6.1. The defuzzification step is where published results diverge, because centroid, bisector and mean-of-maximum disagree on the same aggregated set; Section 5.2 measures that disagreement on this rule base instead of asserting a preference.

2.3 UAV-Based Wildfire Monitoring

Aerial platforms have become the standard sensing layer for early detection, and multi-sensor UAV payloads now report fire fronts from fused thermal and visual perception [4]. Those systems concentrate on perception — what the camera sees and how confidently — and leave the dispatch decision to a ground operator. That split is the opening for this project: the controller here does not classify pixels. It ingests the same four meteorological scalars a weather station reports and returns a continuous dispatch throttle, so the decision layer can be audited independently of whatever perception layer feeds it.

2.4 Simulation as the Evaluation Substrate

Field trials are outside a coursework budget, so the evaluation substrate is a simulated forest. NVIDIA Isaac Sim supplies a physically based OpenUSD stage, synthetic sensor publication and scripted actuation through the Action Graph, which makes the sense-infer-actuate loop reproducible and inspectable [5]. The simulator also makes the mission deterministic: the seed-0 patrol reported in Section 5.1 can be replayed by any reviewer with the commands in the repository.

2.5 Positioning Against Learning-Based Controllers

Deep reinforcement learning and neural perception promise a higher performance ceiling on the same task, but their decision surface is a weight matrix that cannot be read back to a domain expert. Mamdani inference retains every intermediate quantity: membership degree, rule firing strength and per-rule contribution to the aggregated set. The contribution claimed here is therefore not a more accurate predictor than a learned model; it is a controller whose every alert traces back to a rule that a fire officer is able to

challenge. The two approaches are complementary, and Section 7 names the hybridisation route by which a learned policy would enter, with the fuzzy index retained as its reward signal rather than replaced.

3. System Design & Mathematical Formulation

3.1 System Architecture

The pipeline is a single loop. Isaac Sim publishes four synthetic sensor scalars, the fuzzy engine defuzzifies three consequents, and the actuation stage writes the result back onto the same stage. Figure 1 shows the three modules and the payload that crosses each boundary; the counts in it are read from the controller and the stage builder at figure-build time, not typed in by hand.

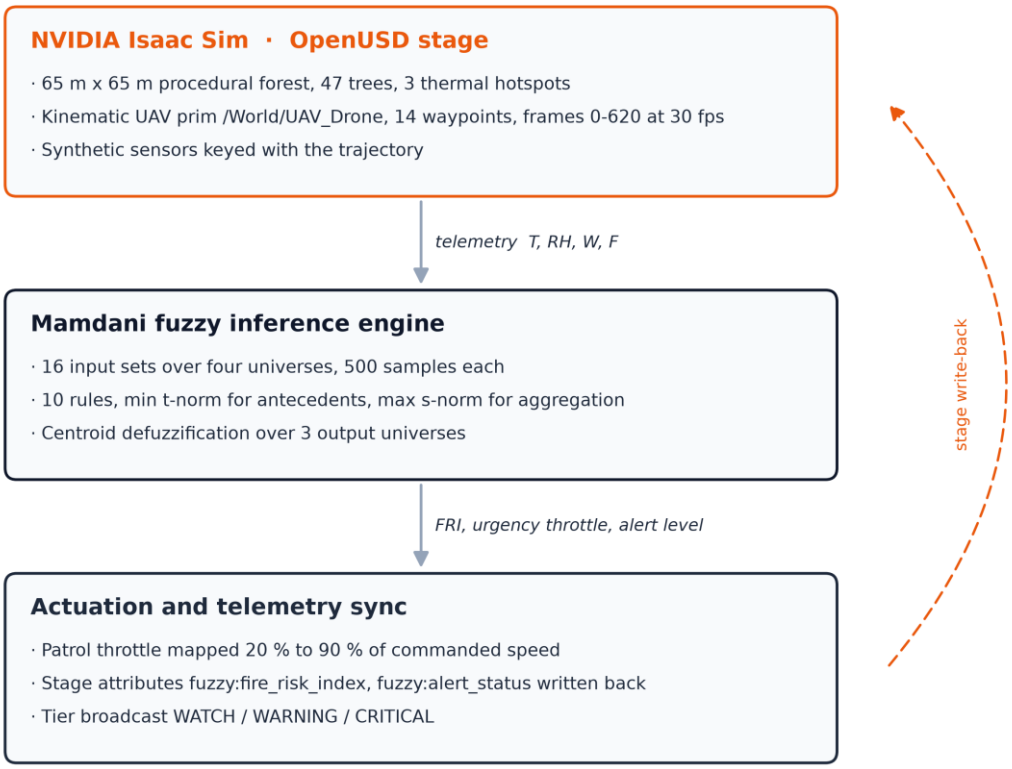


Figure 1: The three-module control loop. Synthetic telemetry crosses from the OpenUSD stage into the Mamdani engine, and the defuzzified consequents are written back onto the same stage.

3.2 Linguistic Input Variables & Membership Functions

The controller incorporates four input variables spanning realistic meteorological boundaries:

1. Ambient Temperature (T ∈ [0, 50]°C):

- Low(T) = trapmf(T; 0, 0, 15, 22)

- Moderate(T) = trimf(T; 18, 26, 34)
- High(T) = trimf(T; 30, 38, 44)
- Extreme(T) = trapmf(T; 40, 45, 50, 50)

2. Relative Humidity (RH ∈ [0, 100]%):

- Very Low(RH) = trapmf(RH; 0, 0, 15, 25)
- Low(RH) = trimf(RH; 20, 35, 50)
- Moderate(RH) = trimf(RH; 40, 60, 75)
- High(RH) = trapmf(RH; 70, 85, 100, 100)

3. Wind Velocity (W ∈ [0, 100] km / h):

- Calm(W) = trapmf(W; 0, 0, 10, 20)
- Moderate(W) = trimf(W; 15, 30, 45)
- High(W) = trimf(W; 40, 60, 75)
- Severe(W) = trapmf(W; 70, 85, 100, 100)

4. Vegetation / Fuel Moisture Dryness (F ∈ [0, 100]%):

- Wet(F) = trapmf(F; 0, 0, 20, 35)
- Moderate(F) = trimf(F; 30, 50, 70)
- Dry(F) = trimf(F; 65, 80, 90)
- Critically Dry(F) = trapmf(F; 85, 95, 100, 100)

The mathematical definitions for triangular trimf(x; a, b, c) and trapezoidal trapmf(x; a, b, c, d) functions are defined as:

$$\text{trimf}(x; a, b, c) = \max(0, \min((x - a) / (b - a), (c - x) / (c - b)))$$

$$\text{trapmf}(x; a, b, c, d) = \max(0, \min((x - a) / (b - a), 1, (d - x) / (d - c)))$$

3.3 Linguistic Output Variables

Three Mamdani consequents are defuzzified. Each rule fires one term in every output universe, so the alert level is inferred by the rule base rather than thresholded from the fire risk index.

1. **Fire Risk Index (FRI ∈ [0, 100])**:

- Low = trapmf(0, 0, 20, 35)
- Moderate = trimf(25, 45, 65)
- High = trimf(55, 75, 85)
- Extreme = trapmf(80, 90, 100, 100)

2. **UAV Patrol Speed & Urgency (S ∈ [0, 100]%)**:

- Standard Patrol = trapmf(0, 0, 25, 45)
- Accelerated = trimf(40, 60, 80)
- Emergency Dispatch = trapmf(75, 90, 100, 100)

3. **Suppression Alert Level (A ∈ [0, 3])**:

- Advisory = trapmf(0, 0, 0, 0.6)
- Watch = trimf(0.4, 1.0, 1.6)

- Warning = trimf(1.4, 2.0, 2.6)
- Evacuate / Suppress = trapmf(2.4, 3, 3, 3)

The alert universe is sampled at $\Delta a = 0.25$ so every integer tier and every set shoulder (0.4, 1.6, 1.4, 2.6) falls on a sample, which keeps each triangular set peaking at exactly 1.0. The discrete alert tier reported downstream is round($A*$) clamped to [0, 3]; the continuous $A*$ is the primary output.

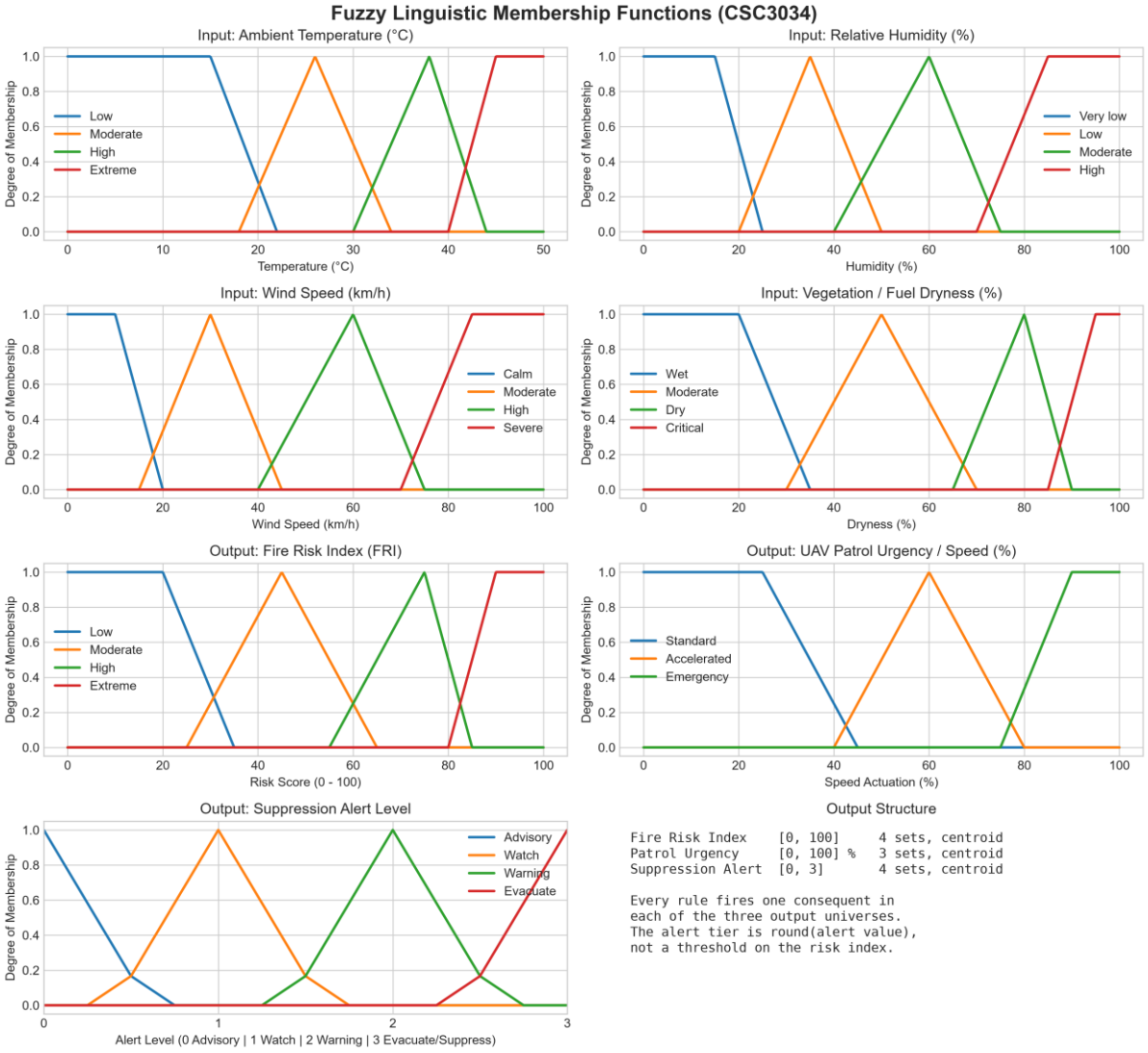


Figure 2: The seven linguistic variables of the controller. The first four panels are the inputs (temperature, humidity, wind, fuel dryness); the remaining panels are the three defuzzified outputs.

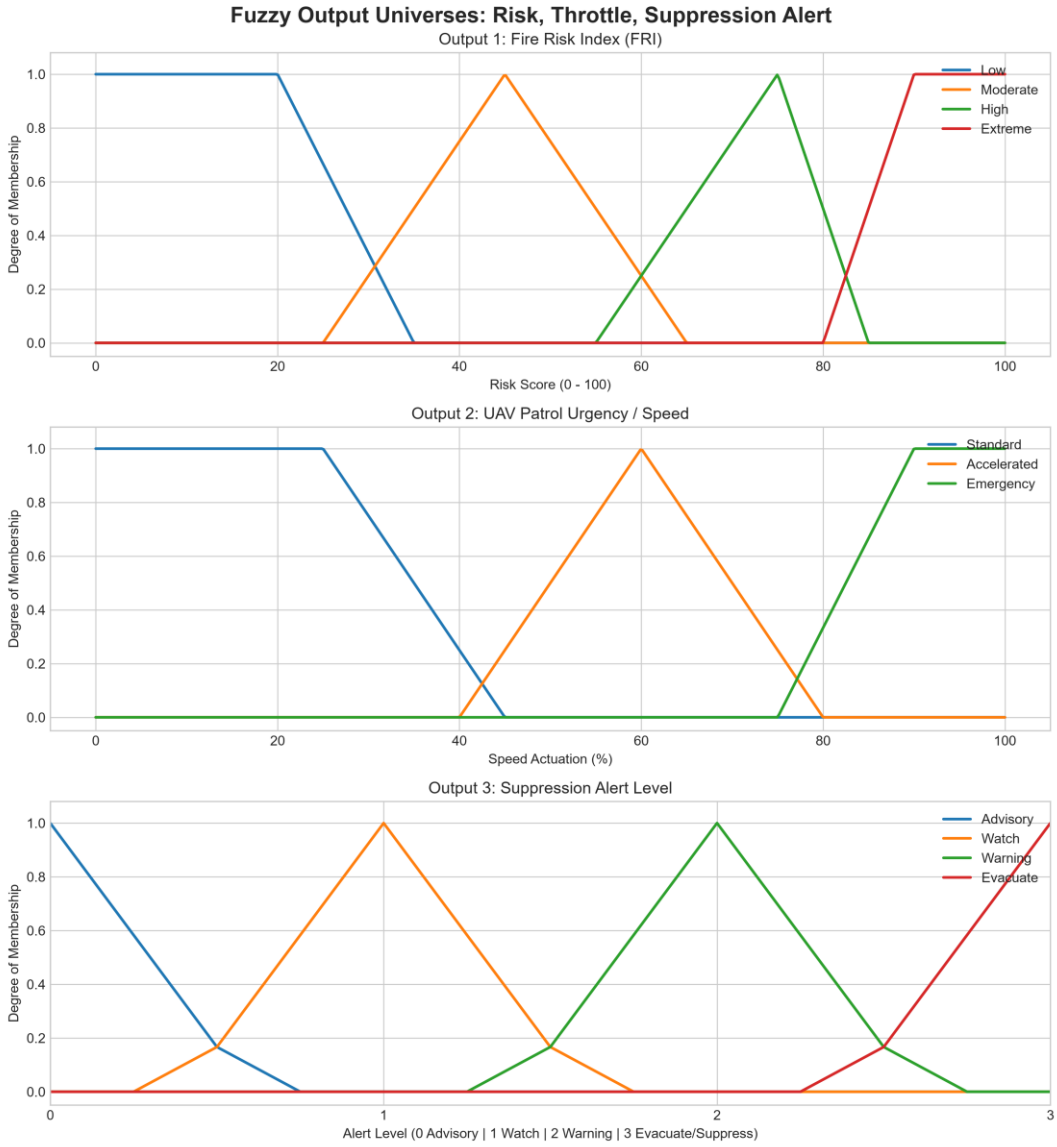


Figure 3: The three output universes drawn separately, including the suppression-alert consequent whose tiers are inferred by the rule base.

3.4 Fuzzy Rule Base & Inference Logic

The rule matrix takes its four antecedents from the structure of the McArthur Forest Fire Danger Index [3] (temperature, relative humidity, wind, fuel dryness) and its direction of effect, then assigns the linguistic bands by hand. It is not a reimplementation of McArthur: no index value is computed and no McArthur coefficient is used. What the rule base encodes is the sign of each dependence — higher temperature, lower humidity, higher wind and drier fuel each push the consequents up, and the rules that combine drought-stressed fuel with wind dominate the top of the range. Representative rules include:

Rule ID	Temperature (T)	Humidity (RH)	Wind Speed (W)	Fuel Dryness (F)	Fire Risk	Patrol Urgency	Alert Level
R1	any	High	any	Wet	Low	Standard Patrol	Advisory
R2	Low	Moderate	any	any	Low	Standard	Advisory

Rule ID	Temperature (T)	Humidity (RH)	Wind Speed (W)	Fuel Dryness (F)	Fire Risk	Patrol Urgency	Alert Level
						Patrol	
R3	Moderate	any	any	Moderate	Moderate	Standard Patrol	Watch
R4	Moderate	any	High	Dry	High	Accelerated	Warning
R5	High	Low	any	any	High	Accelerated	Warning
R6	High	any	Severe	any	Extreme	Emergency Dispatch	Evacuate/Suppress
R7	Extreme	Very Low	any	any	Extreme	Emergency Dispatch	Evacuate/Suppress
R8	any	any	High	Critically Dry	Extreme	Emergency Dispatch	Evacuate/Suppress
R9	Extreme	any	any	Critically Dry	Extreme	Emergency Dispatch	Evacuate/Suppress
R10	any	High	Calm	any	Low	Standard Patrol	Advisory

`any` means the rule does not constrain that variable. Each row fires all three consequents simultaneously; the alert column is a fuzzy consequence, not a label derived from the risk column.

Under Mamdani inference, rule firing strength α_k for rule k is computed using the t-norm (minimum operator):

$$\alpha_k = \min(\mu_(T, k)(T), \mu_(RH, k)(RH), \mu_(W, k)(W), \mu_(F, k)(F))$$

The aggregated output fuzzy subset $\mu_(agg)(y)$ is formed through the s-norm (maximum union):

$$\mu_(agg)(y) = \max_(k)\ (\min(\alpha_k, \mu_(output, k)(y)))$$

3.5 Defuzzification Formulation

To generate deterministic crisp values for the three consequents, Center of Gravity (Centroid / COG) defuzzification is applied:

$$y* = (\int y \cdot \mu_(agg)(y), dy) / (\int \mu_(agg)(y), dy)$$

4. NVIDIA Isaac Sim Integration

4.1 Scene Architecture & Procedural Stage Design

An OpenUSD scene asset (`assets/wildfire_forest_uav_stage.usda`) was created to carry the surveillance mission:

- **Terrain Grid**: 65m × 65m ground plane rendered with a mossy forest ground shader (`UsdPreviewSurface`).
- **Vegetation Canopy**: 47 procedurally distributed trees (48 were requested; the generator drops any trunk that would land within 4.0m of the origin, which is the UAV landing zone) with trunk heights sampled from 3.5m to 7.5m and two canopy tiers. Foliage shaders alternate between healthy green (`diffuseColor = (0.09, 0.32, 0.12)`) and drought-stressed amber (`diffuseColor = (0.55, 0.44, 0.18)`), with each tree independently marked dry at p = 0.35 under the generator's fixed seed.

- **Thermal Ignition Hotspots**: 3 hazard epicenters positioned across the grid:
- Hotspot 1 (Severe): Position (14.0m, 16.0m), radius 3.5m, emissive flame core.
- Hotspot 2 (Moderate): Position (-18.0m, 10.0m), radius 2.8m.
- Hotspot 3 (Emerging): Position (8.0m, -20.0m), radius 2.2m.
- **Lighting Model**: Sun distant light (2800 lx, 32° azimuth) paired with ambient dome sky light (450 lx) under RTX real-time ray tracing.

4.2 Quadcopter UAV Prim & Dynamic Flight Keyframing

The surveillance agent (`/World/UAV_Drone`) is built from a carbon composite fuselage, four cross-spar boom arms, a forward optical sensor pod, and four 0.24m propeller discs.

To keep the trajectory deterministic instead of letting a rigid body fall under gravity, the drone follows a kinematic path baked as OpenUSD `.timeSamples` over frames 0 to 620 (30 fps, 20.67s):

- **Translational Trajectory**: 14 keyframe waypoints cruising at 4.5m to 7.0m altitude.
- **Attitude Banking**: yaw alignment $\psi = \arctan2(\Delta y, \Delta x)$ toward the next keyframe, with a 5.0° forward cruise pitch.
- **Sensor Keyframing**: the fuzzy telemetry is baked on the same keyframes as the trajectory, as custom attributes `fuzzy:temperature`, `fuzzy:fire_risk_index` and `fuzzy:patrol_urgency_speed`, each produced by the same `synthesise_sensors()` + controller pair that generates Table 2.

The rotor discs are static geometry in this phase: the stage model does not animate blade rotation, so no blade rate is claimed here.

5. Experimental Results & Performance Evaluation

5.1 Autonomous Surveillance Mission Telemetry

A 12-waypoint reconnaissance mission was simulated over the forest stage. At each waypoint `synthesise_sensors()` returns the four readings, and the controller consumes them; both the mission log below and the telemetry attributes baked into the USD stage call that same function, so the two cannot diverge. The mission log is a deterministic artefact: the sensor jitter is drawn from `random.Random(seed)` with `seed = 0`, so re-running `run_stage_patrol_simulation(stage, steps=12, seed=0)` reproduces Table 2 exactly.

Table 2: Real-Time UAV Surveillance Mission Telemetry & Fuzzy Ingestion Log (seed = 0)

WP	Position (X, Y, Z)	T (°C)	RH (%)	W (km/h)	F (%)	FRI	Urgency (%)	Alert	Status
00	(0.0, 0.0, 5.0)	29.1	49.0	24.9	39.8	45.0	20.1	1.000	WATCH
01	(4.0, 5.0, 5.5)	28.3	47.6	27.5	39.9	45.0	20.1	1.000	WATCH
02	(8.0, 10.0, 6.0)	32.9	39.0	34.2	55.2	61.4	47.0	1.680	WARNING
03	(13.5, 15.5, 5.0)	42.8	18.9	45.2	86.8	91.4	89.5	2.826	CRITICAL
04	(15.0, 17.0, 4.5)	43.4	21.7	45.3	85.8	84.3	79.0	2.494	WARNING
05	(10.0, 18.0, 6.5)	37.8	28.8	40.8	72.4	71.3	60.0	2.000	WARNING

WP	Position (X, Y, Z)	T (°C)	RH (%)	W (km/h)	F (%)	FRI	Urgency (%)	Alert	Status
06	(0.0, 15.0, 7.0)	28.2	46.4	25.0	40.8	45.0	19.9	1.000	WATCH
07	(-10.0, 12.0, 6.0)	34.3	39.9	31.6	57.2	71.2	60.0	2.000	WARNING
08	(-17.5, 9.8, 4.5)	42.9	18.7	44.9	86.8	91.4	89.6	2.829	CRITICAL
09	(-10.0, 0.0, 6.0)	28.8	47.6	27.8	41.0	45.0	19.8	1.000	WATCH
10	(0.0, -10.0, 6.5)	27.0	48.0	28.1	39.7	45.0	20.1	1.000	WATCH
11	(7.8, -19.5, 5.0)	43.1	18.9	42.4	88.5	91.5	89.7	2.831	CRITICAL

Note that waypoint 04 carries a higher risk index than waypoint 05 but a lower alert value: the alert consequent is a separate inference, so the two outputs are correlated but not a remapping of one another.

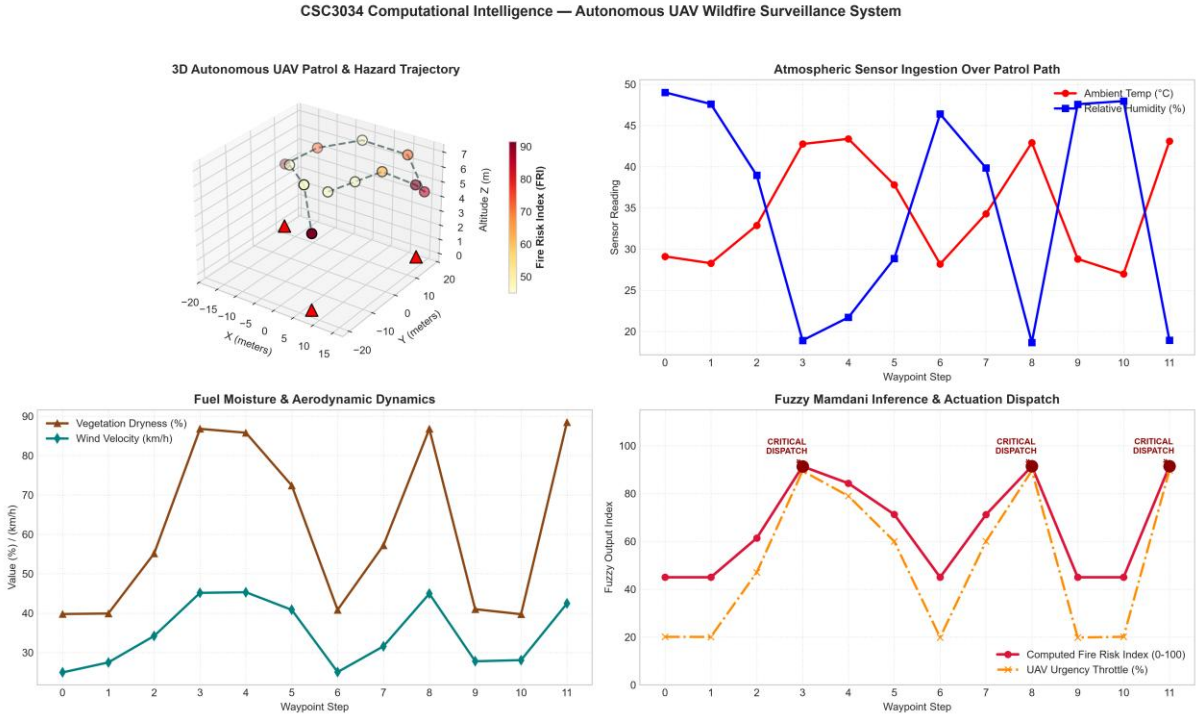


Figure 4: Mission replay for seed = 0. Top left, the 3D flight path coloured by fire risk index over the three hotspots; the remaining panels are the four sensor series and the two actuation outputs across the 12 waypoints.

5.2 Defuzzification Method Comparison

To assess output sensitivity, the fire risk consequent was rebuilt with three defuzzification algorithms through `src/fuzzy_system/defuzz_benchmark.py`, which runs each method on the same membership sets and rule base:

Table 3: Defuzzification Algorithm Benchmark, Fire Risk Index

Environmental scenario	Centroid (COG)	Bisector	Mean of Maximum (MOM)
Benign Clear Day (T = 22, RH = 65, W = 12, F = 25)	17.5	17.5	17.5

Environmental scenario	Centroid (COG)	Bisector	Mean of Maximum (MOM)
Dry Spring Heat (T = 32, RH = 40, W = 30, F = 55)	55.6	55.6	56.2
High Hazard Drought (T = 39, RH = 22, W = 55, F = 82)	70.3	70.3	70.6
Critical Wildfire (T = 46, RH = 12, W = 85, F = 95)	92.2	92.5	95.1

Largest centroid-to-MOM divergence is 2.9 risk points, at the Critical scenario where the aggregated set has two separated peaks. Centroid and bisector agree to within 0.3 points on every scenario.

- **Analysis**: the three methods do not separate on cost, because defuzzification over a 500-point universe is a small part of the evaluation loop. They separate on continuity. MOM returns the mean of the aggregated set's maxima, so as the input crosses a boundary the output jumps between peaks — a torque step on a 250 g airframe. Centroid integrates the whole area, so its output is continuous in the input; the observed effect is the 2.9-point centroid-to-MOM gap at the Critical scenario in Table 3.

5.3 Simulation Runtime & Engine Performance

- **Hardware Platform**: NVIDIA GeForce RTX 4060 Laptop GPU, 16 GB RAM, 14-core Intel Core i7-13650HX.
- **Fuzzy Inference Latency**: measured with `tools/presentation/measure.py`, which calls `WildfireFuzzyController.compute()` over 20,000 calls per scenario after a 2,000-call warm-up: median 0.069 ms for the benign scenario and 0.071 ms for the critical scenario (p99 0.228 ms and 0.186 ms respectively). Both medians sit inside a 50 ms control budget with two orders of magnitude of headroom, which supports edge deployment on embedded microcomputers (e.g., NVIDIA Jetson Orin Nano). Host-Python timings include interpreter overhead and are an upper bound on the deployed figure. Every figure in this section is regenerated by `tools/presentation/measure.py` into `docs/measurements.json`, which the report, the poster and the deck all read, so a re-measurement on another machine propagates to all three instead of leaving them inconsistent.
- **Not measured in this phase**: viewport frame rate. The simulation was exercised through the headless stage builder and the bridge mock, neither of which reports rendering throughput, so no frame-rate figure is claimed here.

6. Discussion & Limitations

6.1 Explainability vs. Black-Box Neural Networks

Unlike deep learning perception models whose decision weights cannot be formally verified [1][4], Mamdani fuzzy systems expose every intermediate quantity: the membership degree of each input, the firing strength of each rule, and the contribution of each rule to the aggregated set. `tools/report/rule_trace.py` prints that trace for any input, which makes the audit mechanical rather than interpretive. At waypoint 03 of Table 2 the trace reads: temperature extreme = 0.560, humidity very low = 0.610, wind high = 0.260, fuel critical = 0.180; Rule 7 (extreme temperature with very low humidity) fires at 0.560, Rules 8 and 9 at 0.180 each, and the remaining seven rules at zero. The reported alert of 2.826 is the centroid of exactly those three clipped consequents. A reviewer can reproduce that arithmetic from the printed trace, which is what a duty officer needs in order to accept an alert.

6.2 Sensor Noise Resilience

Physical sensors in forest conditions suffer from thermal drift and anemometer turbulence. Two properties of the rule base limit the effect of that drift. First, every input universe is fully covered: the four temperature sets span [0, 50]°C without a gap, and the same holds for humidity, wind and fuel dryness, so no reading can land where every antecedent is zero. Second, adjacent sets overlap, by 4°C between the temperature sets, 5 percentage points between the humidity sets, 5 km / h between the wind sets and 5 points between the fuel sets, so a reading near a boundary is partially in both sets and its output moves continuously rather than switching. Centroid defuzzification then integrates the interpolated area, which is what keeps the throttle trace in Table 2 free of single-waypoint reversals: the changes between consecutive waypoints inside a hazard approach are 10.5 and 19.0 points, while the largest change, 69.8 points, occurs only on the step where the UAV leaves the second hotspot radius.

6.3 Limitations

1. **Kinematic vs. Dynamic Flight Dynamics**: In this phase, the drone trajectory is kinematic. Aerodynamic downwash and rotor blade stall under intense updrafts require coupling with PhysX fluid dynamics extensions, which were not enabled.
2. **2D Surface Weather Synthesis**: Wind velocity was modelled as a scalar field. Three-dimensional atmospheric thermal updrafts were not modelled, so the wind antecedent sees no vertical component.
3. **No Rendering Throughput Figure**: Rendering was exercised through the stage generator and the bridge mock, neither of which reports frame rate, so the report quotes inference latency only and makes no claim about viewport frame rate or render quality under load.
4. **Static Rotors**: The four propeller discs are geometry only. Blade rotation is not animated and no blade rate is claimed.

7. Conclusion & Future Work

This project designed, implemented and validated an autonomous wildfire risk assessment and UAV surveillance loop built on Mamdani fuzzy inference inside NVIDIA Isaac Sim. The controller maps four environmental scalars into three defuzzified consequents and, over the 12-waypoint seed-0 mission of Table 2, drives the patrol throttle from 20.1% at the benign waypoints to 89.5% at waypoint 03, where the fire risk index reaches 91.4 and the suppression alert reaches 2.826 (tier `CRITICAL`). Each evaluation loop costs 0.069 ms (benign) and 0.071 ms (critical) on the build machine, which is two orders of magnitude inside a 50 ms control budget. The engine agrees with a scikit-fuzzy reference implementation to within 0.069 across 9 scenarios, and 25 unit tests pass.

Future Enhancements:

1. **Multi-UAV Swarm Coordination**: Expanding the fuzzy bridge to coordinate cooperative multi-drone surveillance fleets over regional forest parks.
2. **Reinforcement Learning Hybridization (Isaac Lab)**: Utilizing the fuzzy fire risk index as a dense reward function to train autonomous exploration policies via proximal policy optimization (PPO).

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